Recap – dealing with LLE

Liquid-liquid separation occurs at a particular temperature when two points exist at which $\frac{d^2 G}{dx_1^{\,2}} = 0$

Gibbs energy is then minimised by “unmixing” into two separate liquid phases

“Normal” behaviour is for the tendency to mix to increase with temperature (due to the ideal entropic term), but the opposite behaviour is also possible

Consolute points are the maximum or minimum temperature (and corresponding composition) at which phase separation occurs

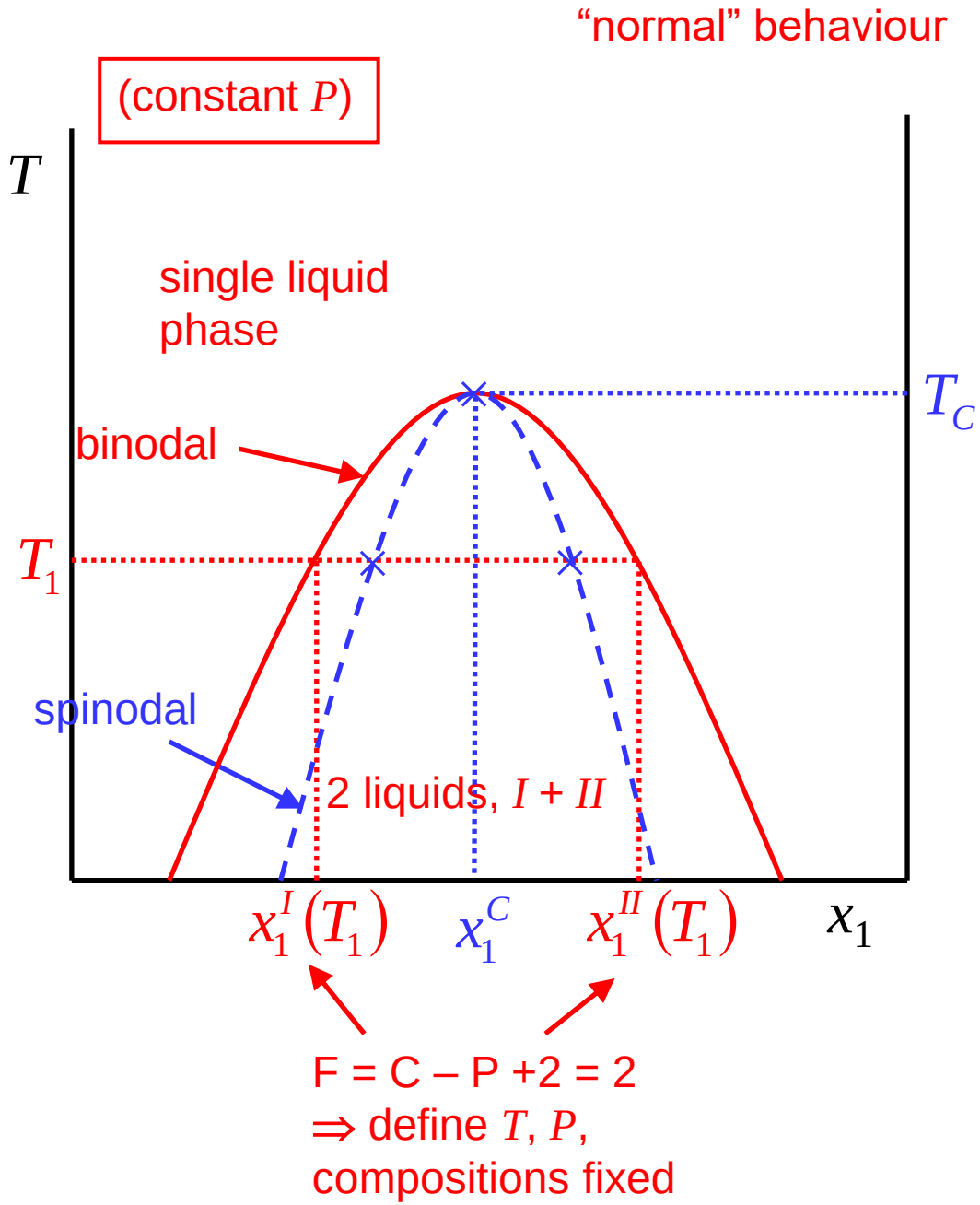
Significance of Binodal and Spinodal Lines

The binodal is the phase boundary between the region in which the equilibrium state is a single liquid phase and that in which two liquid phases of different composition are the equilibrium state

The spinodal is the line connecting points at which $\left(\frac{\partial^2 G}{\partial x_1^{\,2}}\right)_{T,P} = 0$

for each temperature for which there are two liquid phases at equilibrium there are two such points

Consider the phase diagram (at constant pressure):


"normal" behaviour
(constant P)
T
single liquid phase
T_C
binodal
T_1
spinodal
2 liquids, I + II
$x_1^I(T_1)$
x_1^C
$x_1^{II}(T_1)$
x_1
F = C – P +2 = 2
⇒ define T, P, compositions fixed

Now lets look at some significant points on a $\underline{G} - x_1$ plot (at constant temperature (and pressure)):

$$\mu_1 = \underline{G} + x_2 \left(\frac{\partial \underline{G}}{\partial x_1} \right)_{T,P} \quad \text{and} \quad \mu_2 = \underline{G} - x_1 \left(\frac{\partial \underline{G}}{\partial x_1} \right)_{T,P}$$

$$\left(\frac{\partial \mu_1}{\partial x_1} \right) = x_2 \left(\frac{\partial^2 \underline{G}}{\partial x_1^{\,2}} \right)_{T,P} \quad \text{and} \quad \left(\frac{\partial \mu_2}{\partial x_1} \right) = -x_1 \left(\frac{\partial^2 \underline{G}}{\partial x_1^{\,2}} \right)_{T,P}$$

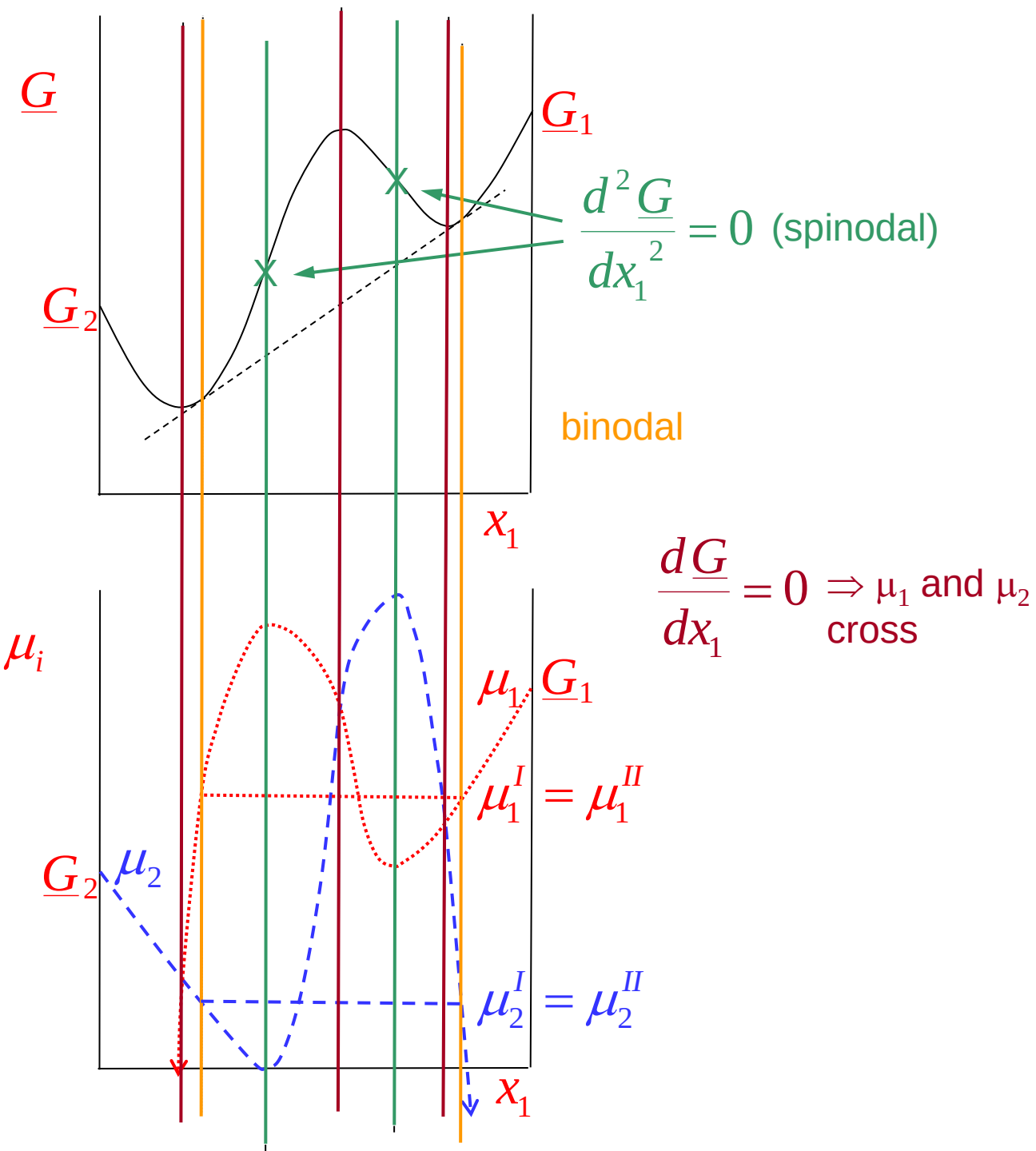
At $x_2 = 0$ $\quad \mu_1 = \underline{G}_1$ and $\quad \mu_2 = -\infty$

At $x_1 = 0$ $\quad \mu_2 = \underline{G}_2$ and $\quad \mu_1 = -\infty$

At points where $\left(\frac{\partial \underline{G}}{\partial x_1} \right)_{T,P} = 0$ $\quad \mu_1 = \mu_2 = \underline{G}$

At points where $\left(\frac{\partial^2 \underline{G}}{\partial x_1^{\,2}} \right)_{T,P} = 0$ $\quad \left(\frac{\partial \mu_1}{\partial x_1} \right) = \left(\frac{\partial \mu_2}{\partial x_1} \right) = 0$

note: between the spinodals μ_1 decreases with increasing x_1 (and μ_2 decreases with increasing x_2)



Chemical potential is the driving force for the movement of material

"Normally“ $\left(\frac{\partial \mu_1}{\partial x_1}\right)_{T,P}$ and $\left(\frac{\partial \mu_2}{\partial x_2}\right)_{T,P}$ are both positive

Thus component (1) will be driven by the chemical potential from a region of higher x_1 to a region of lower x_1

This is stable, because the movement of material acts to remove the gradient of mole fraction.

Now consider mole fractions between the spinodal points

$\left(\frac{\partial \mu_1}{\partial x_1}\right)_{T,P}$ and $\left(\frac{\partial \mu_2}{\partial x_2}\right)_{T,P}$ are both <u>negative</u>

This means that as component (1) moves down a chemical potential gradient it moves from a region of lower x_1 to a region of higher x_1

This is clearly an unstable situation, as the mole fraction gradient is increased by the movement of material

The region between the binodal and spinodal lines is one in which the equilibrium state is two liquid phases (of different composition) simultaneously present

However it is possible for a single liquid phase to be present as a metastable state

In order for phase separation to occur in this region, we require some sort of nucleation site

Between the two spinodal lines, a single phase is unstable, because material spontaneously moves up a mole fraction gradient

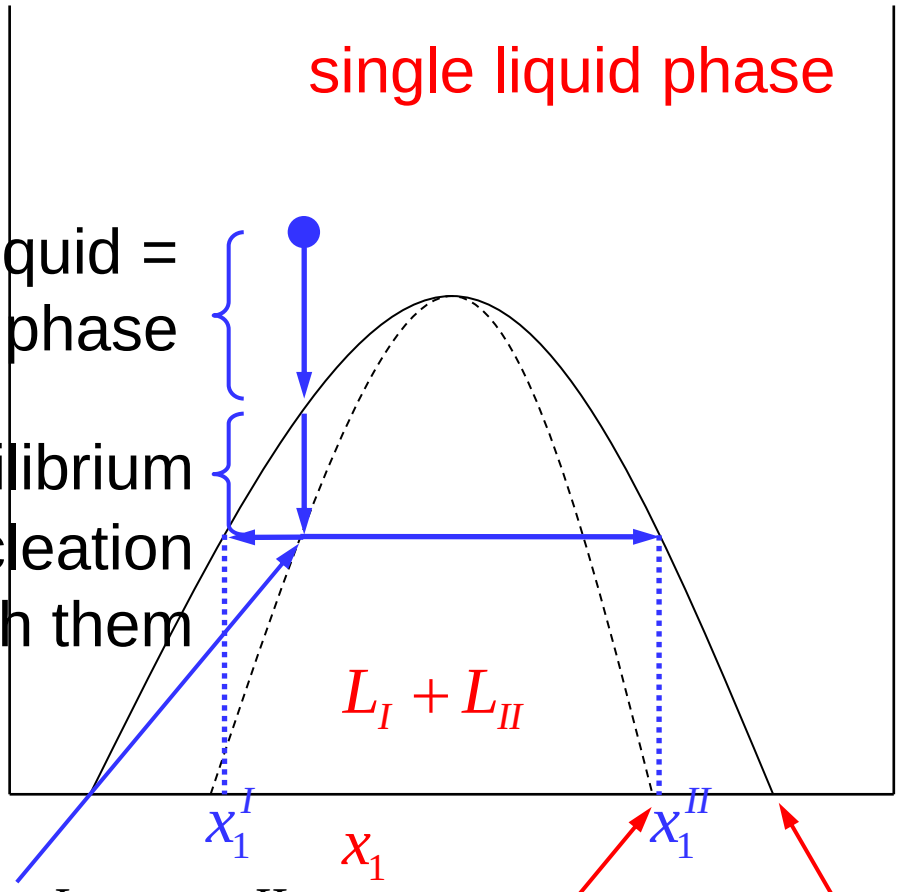
It is impossible ever to achieve a single liquid phase in this region (even temporarily)

Once we cross the spinodal line, liquid phases I and II spontaneously separate even in the absence of nucleation sites

This results in a very fine dispersion of liquid phase I in liquid phase II (or vice versa)

This is called spinodal decomposition

Consider cooling from a stable single liquid phase:



single liquid = equilibrium phase

$L^I + L^{II}$ are equilibrium phases, but nucleation required to reach them

At spinodal, L^I and L^{II} separate spontaneously

spinodal $\frac{d^2 G}{dx_1^2} = 0$

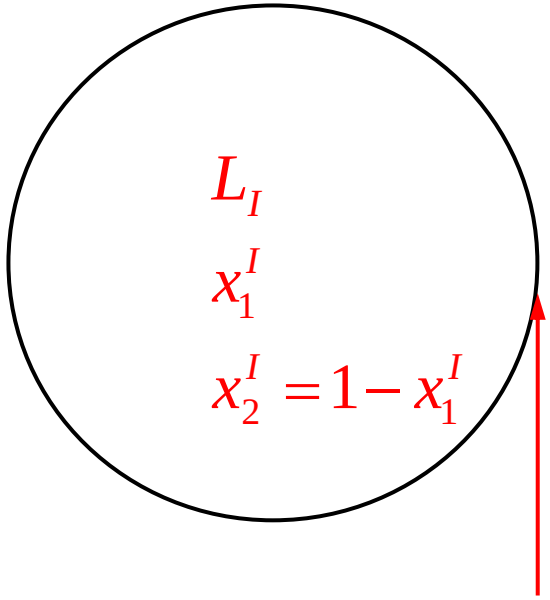
binodal = equilibrium phase boundary

Diffusion in LLE

We normally write the diffusion equation in a form such as:

$$N_1 = -Dc\frac{dx_1}{dz}$$

But consider the boundary between liquid phases I and II at equilibrium:



L_I

x_1^I

$x_2^I = 1 - x_1^I$

L_{II}

x_1^{II}

$x_2^{II} = 1 - x_1^{II}$

$x_1^I \neq x_1^{II}$

At boundary $\frac{dx_1}{dz} = \infty$

How can this be?

In reality material moves down a gradient in chemical potential, not in mole fraction

Consider a binary Ideal Mixture

$$\mu_1 = \underline{G}_1 + RT \ln x_1$$

$$\left(\frac{\partial \mu_1}{\partial x_1}\right)_{T,P} = \frac{RT}{x_1}$$

and so we can write, $N_1 = -Dc\frac{dx_1}{dz} = -Dc\left(\frac{\partial x_1}{\partial \mu_1}\right)\frac{d\mu_1}{dz} = -\frac{Dcx_1}{RT}\frac{d\mu_1}{dz}$

Fick's law

change variable

For an IM we can write a satisfactory Fickean diffusion equation using the gradient in chemical potential as driving force

Now consider the binary LLE case

Far from being an Ideal Mixture – else mixing would be perfect

Consider the spinodal:

$$\left(\frac{\partial^2 G}{\partial x_1^{\,2}}\right) = \left(\frac{\partial \mu_1}{\partial x_1}\right) = 0$$

There can be no diffusion, since there is no chemical potential gradient

If we try to measure a conventional diffusion coefficient, we will find it is zero at the spinodal composition!

This can be observed experimentally very close to the consolute point

Consider the binodal: at the $L^I - L^{II}$ phase boundary, we have $\mu_1^I = \mu_1^{II}$ and $\mu_2^I = \mu_2^{II}$ and so again there is no diffusion because there is no chemical potential gradient (although there is a very steep mole fraction gradient)

Between the spinodals: $\left(\frac{\partial \mu_1}{\partial x_1}\right)_{T,P}$ and $\left(\frac{\partial \mu_2}{\partial x_2}\right)_{T,P}$ are both <u>negative</u>

and so diffusion occurs <u>up</u> the mole fraction gradient

Consider a real binary mixture

$$\underline{G} = \underline{G}^{IM} + \underline{G}^{ex} = \underline{G}^{IM} + RTx_1 \ln\gamma_1 + RTx_2 \ln\gamma_2$$

$$\mu_1 = \underline{G}_1 + RT \ln x_1 + RT \ln\gamma_1$$

$$\left(\frac{\partial\mu_1}{\partial x_1}\right)_{T,P} = RT\left(\frac{1}{x_1} + \frac{\partial \ln\gamma_1}{\partial x_1}\right)_{T,P}$$

and so, by analogy with the IM case, we would like to write:

assume this really is a constant

$$N_{11} = -\frac{Dcx_1}{RT}\frac{d\mu_1}{dz} = -\frac{Dcx_1}{RT}\left(\frac{\partial\mu_1}{\partial x_1}\right)\frac{dx_1}{dz}$$

$$= -Dcx_1\left(\frac{1}{x_1} + \frac{\partial \ln\gamma_1}{\partial x_1}\right)\frac{dx_1}{dz} = -Dc\left(1 + \frac{\partial \ln\gamma_1}{\partial \ln x_1}\right)\frac{dx_1}{dz}$$

(changing back to concentration gradient as the driving force)

ideal term

non-ideal contribution

In fact to fit experiments close to consolute points in LLE situations (i.e. where $\frac{\partial \mu_1}{\partial x_1} \approx 0$), we require an equation of the form:

$$N_1 = -Dc\left(1+\frac{\partial \ln \gamma_1}{\partial \ln x_1}\right)^{n}\frac{dx_1}{dz}$$

Recent publications [1,2,3,4] show that a value of n of ~0.6 works well

1. C. D'Agostino, M.D. Mantle, L.F. Gladden and G.D. Moggridge, Chem. Eng. Sci. 66 (2011) 3898-3906.

2. G.D. Moggridge, Chem. Eng. Sci. 71 (2012), 226-238.

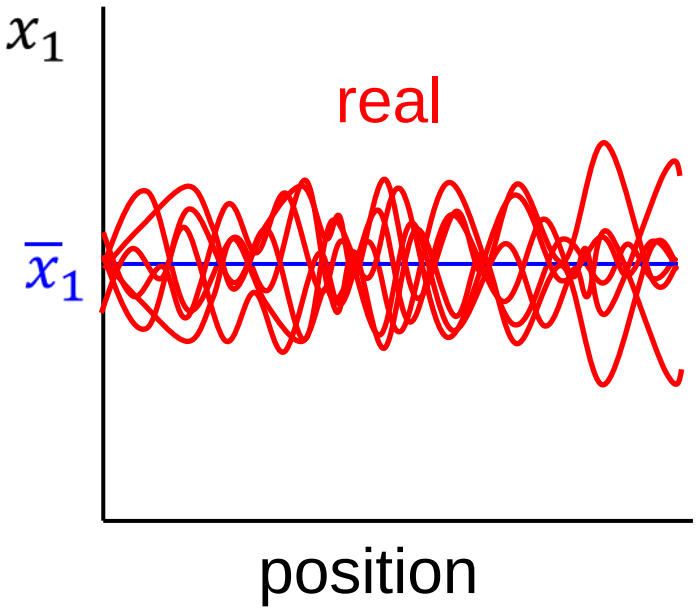
3. C. D'Agostino, M.D. Mantle, L.F. Gladden and G.D. Moggridge, Chem. Eng. Sci. 74 (2012) 105-113.

4. G.D. Moggridge, Chem. Eng. Sci. 76 (2012), 199-205.

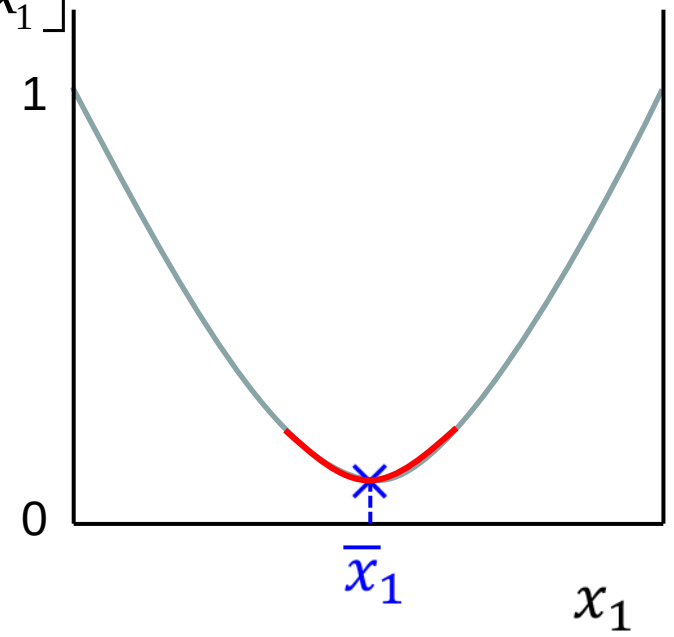
$$N_1 = -Dc\left[1+\frac{d\ln\gamma_1}{d\ln x_1}\right]^n \frac{dx_1}{dz}$$

Why is *n* not equal to one?

Dynamic concentration fluctuations

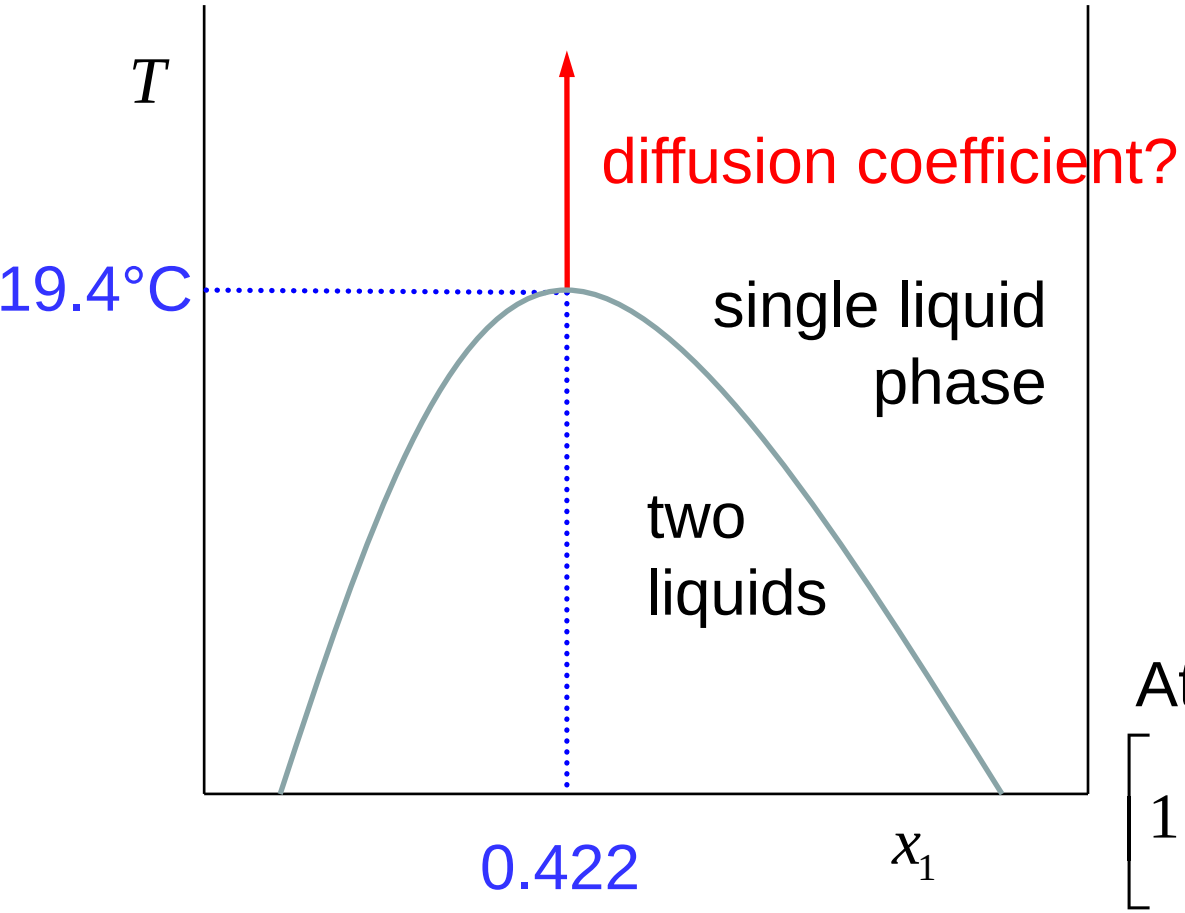


$$\left[1+\frac{d\ln\gamma_1}{d\ln x_1}\right]$$



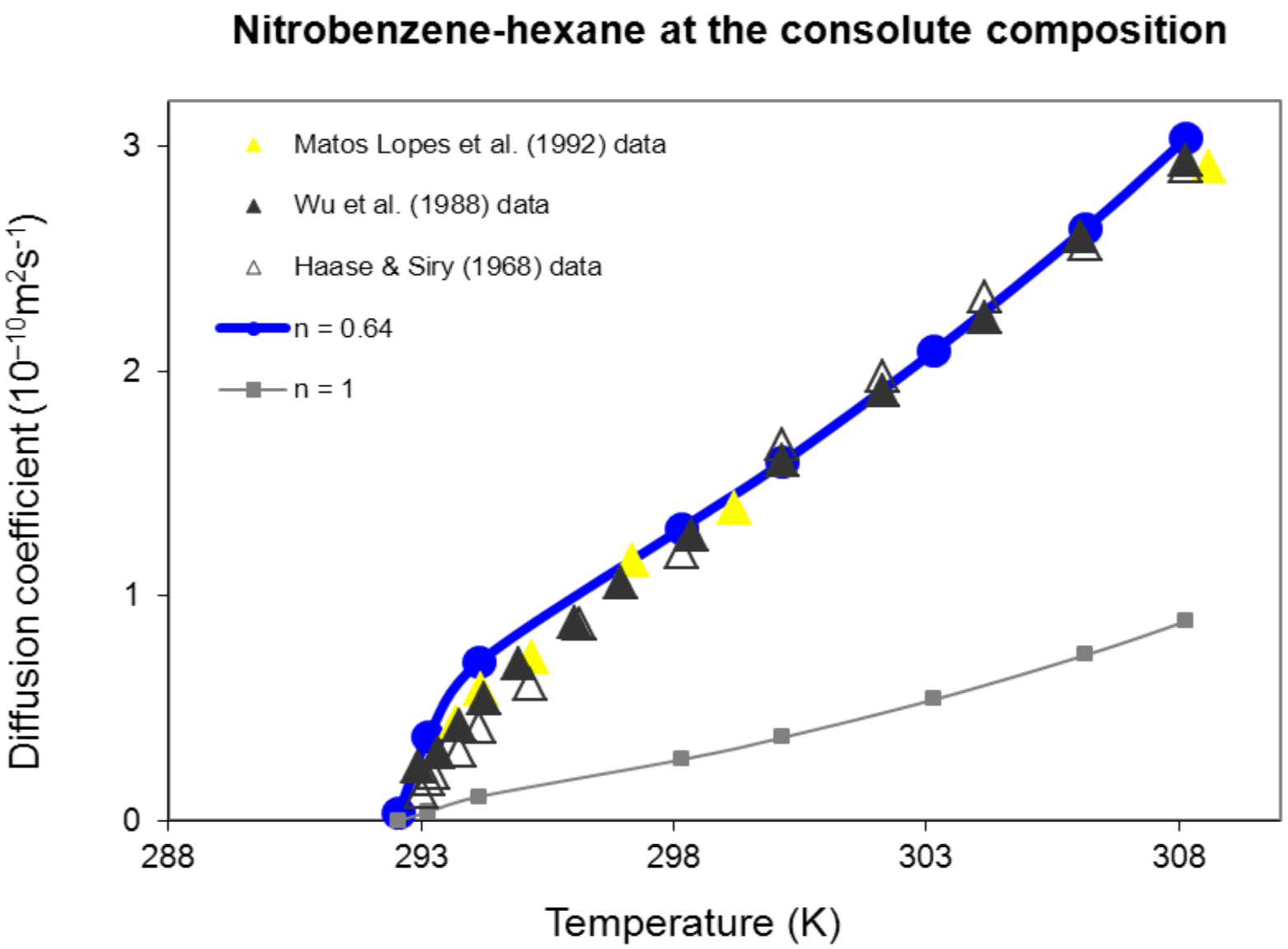
Example

Nitrobenzene (1) and hexane (2)



At the consolute point

$$\left[1+\frac{d\ln\gamma_1}{d\ln x_1}\right]=0 \Rightarrow D_{12}=0$$


Nitrobenzene-hexane at the consolute composition
Matos Lopes et al. (1992) data
Wu et al. (1988) data
Haase & Siry (1968) data
n = 0.64
n = 1
Diffusion coefficient (10⁻¹⁰m²s⁻¹)
Temperature (K)
0
1
2
3
288
293
298
303
308

How was $\left[1+\frac{d\ln\gamma_1}{d\ln x_1}\right]$ calculated?

Using a 4-constant Redlich-Kister expansion activity coeff model

fit to P–x_1 data from Neckel & Volk (1968) at 21°C, 25°C and 35°C

Poynting correction

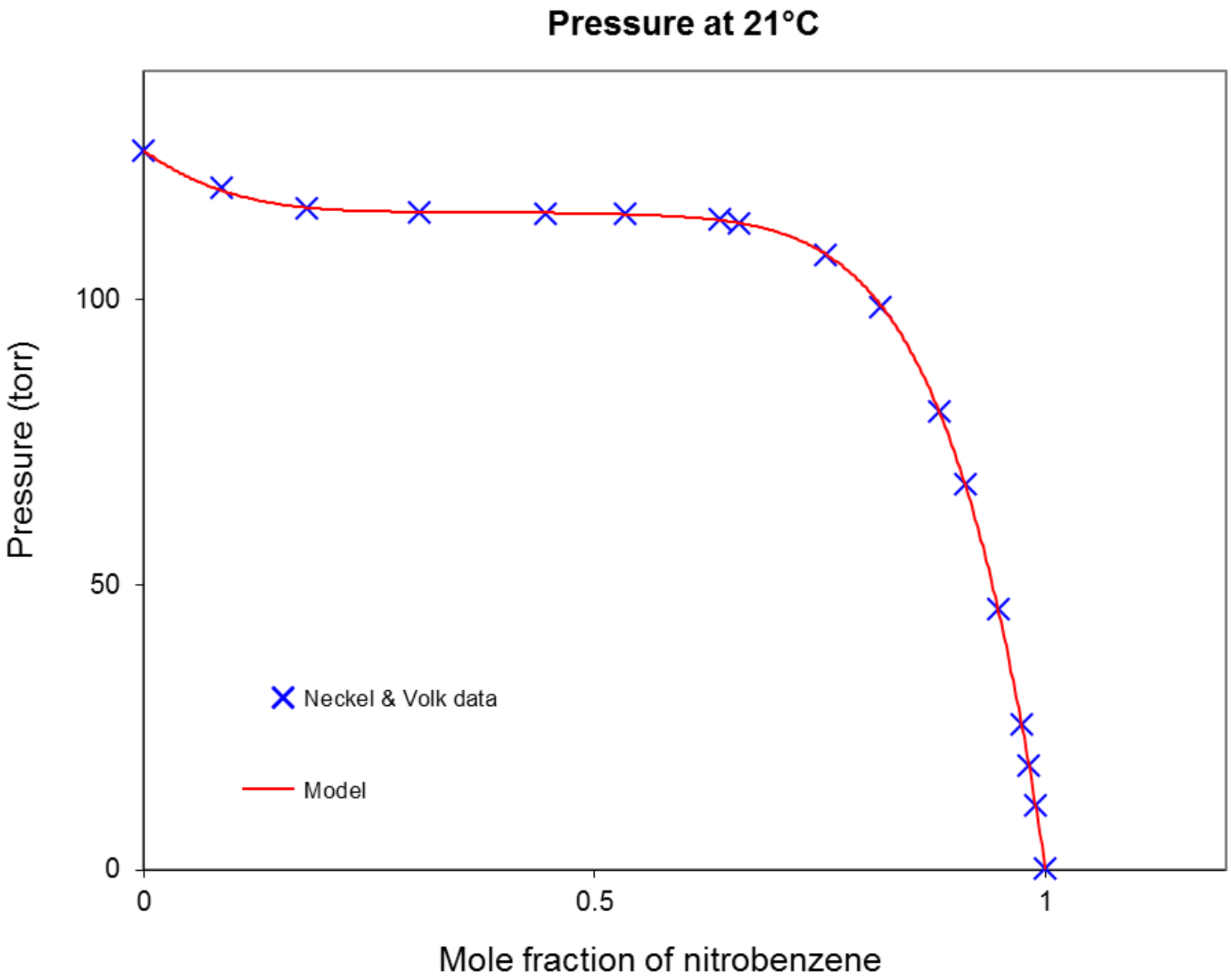
non-ideal vapour correction

$$P = x_1\gamma_1 P_1^{SAT}\exp\left\{\frac{\left(\underline{V}_1^L - B_{11}\right)\left[P - P_1^{SAT}\right]}{RT}\right\} + x_2\gamma_2 P_2^{SAT}\exp\left\{\frac{\left(\underline{V}_2^L - B_{22}\right)\left[P - P_2^{SAT}\right]}{RT}\right\}$$

data

activity coeff model = fn (x_1, 4 parameters)

only unknowns


Pressure at 21°C
Pressure (torr)
100
50
0
0
0.5
1
Mole fraction of nitrobenzene
Neckel & Volk data
Model